Status Report #2

Oct 29, 2026

The machine learning team trained an initial DistilRoBERTa text classifier and tuned its classification threshold. Validation results meet the project targets, but the unusually high threshold needs further investigation.

Reporting period: October 14–29, 2026.

Progress

Tara and Noa completed the first text-classifier training run and evaluated classification thresholds on validation data.

Area	What we did	Status
Model and dataset	Selected **DistilRoBERTa** for text classification and **MAGE** for AI-generated versus human-written text.	Selected
Data preparation	Created scripts to explore, clean, deduplicate, standardize labels, and prepare training, validation, and test splits.	Done
Training setup	Configured PyTorch, Hugging Face Transformers, and Google Colab GPU training.	Done
Initial training	Built the training pipeline and completed a run with **20,000 training examples**.	Done
Validation results	Initial F1: **0.818**; false-positive rate: **41.48%**.	Needs improvement
Threshold tuning	At a **0.999** threshold, validation F1: **0.826** and false-positive rate: **9.3%**; both meet the project targets.	Validation targets met
Documentation	Recorded model choice, data preparation, training, and evaluation results in GitHub.	Done

Next steps

The machine learning track will validate the selected threshold on unseen data before deciding whether to integrate the classifier into RealEye.

1. Evaluate the trained model at the **0.999 threshold** on the held-out MAGE test set.

2. Calculate test accuracy, precision, recall, F1 score, and false-positive rate.

3. Compare test and validation metrics to assess generalization.

4. If test results meet the project requirements, integrate the trained model and threshold into the RealEye backend.

5. If performance falls short, analyze classification errors and consider adjusting training parameters, improving the dataset, calibrating weights (for example, with logistic regression), or comparing other pretrained models.

Troubleshooting and open questions

High false-positive rate: The initial classifier incorrectly flags many human-written examples as AI-generated (41.48% false-positive rate on validation data). Some incorrect predictions receive high confidence scores.

Unusually high threshold: Increasing the threshold to 99.9% lowers the validation false-positive rate to 9.3% while retaining an F1 score of 0.826. Although these results meet our targets, needing such a high cutoff raises questions about the reliability and calibration of the model's confidence scores.

Potential improvements: Should we further fine-tune the model, adjust training settings, calibrate weights using logistic regression, or train another pretrained classifier for comparison? We will use held-out test results and error analysis to guide this decision.